Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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Ringkasan

PENDAHULUAN
TINJAUAN PUSTAKA
amsfonts, amssymb
Proposition R.231: Prove that $A = \left\{m + n\sqrt{3} \mid m, n \in Z\right\}$ is closed under mulitplication.
Proof: Let $A = \left\{m + n\sqrt{3} \mid m, n \in Z\right\}$, and let $m + n\sqrt{3}$ and $p + q\sqrt{3}$ be elements of A. Then

$$\begin{aligned}\left(m + n\sqrt{3}\right)\left(p + q\sqrt{3}\right) &= mp + mq\sqrt{3} + np\sqrt{3} + 3qn \\ &= (mp + 3qn) + (mq + np)\sqrt{3}.\end{aligned}$$

Since $m, n, p, q \in Z$, $mp + 3nq$ and $mq + np$ are both integers. Therefore,

$$\left(m + n\sqrt{3}\right)\left(p + q\sqrt{3}\right) \in A,$$

and A is closed under multiplication.